



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: VI

Subject Name: Pharmaceutical Biotechnology

Subject Code: BP605T

Max. Marks: 75

Date: 25/07/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

i)biosensors are based on the principle of sound vibrations.

- | | |
|------------------|-------------------|
| a) Piezoelectric | b) Optical |
| c) Calorimetric | d) Potentiometric |

ii) Which type of restriction endonucleases is used most in genetic engineering?

- | | |
|-------------|------------|
| a) Type I | b) Type II |
| c) Type III | d) Type IV |

iii) A segment of DNA that reads the same from forward and backward is called

- | | |
|--------------------|----------------------|
| a) palindromic DNA | b) complementary DNA |
| c) plasmid DNA | d) copy DNA |

iv) Which of the following is not included in immobilization process?

- | | |
|---------------|------------------|
| a) Absorption | b) Adsorption |
| c) Entrapment | d) Cross-linking |

v) Most commonly used plasmid vector in genetic engineering is

- | | |
|------------|------------|
| a) pBR 322 | b) pBR 328 |
| c) pBR 232 | d) pBR 832 |

vi) Name the class of immunoglobulin which has a pentameric structure?

- | | |
|---------|---------|
| a) IgE | b) Ig G |
| c) Ig A | d) IgM |

vii) Restriction endonucleases are utilised in genetic engineering as.....



- a) Molecular build up at nucleotides
- b) Molecular degradation to DNA break up
- c) Molecular cement for combining DNA
- d) Molecular scalpels for cutting DNA

viii) What will be the consequence of not having an origin of replication (ori) in the vector?

- a) If an ori is absent, replication of vector would not take place
- b) As the cells divide after taking up the vector, both the daughter cells would be having the vector
- c) A colony of transformed colonies is observed
- d) The vector won't be taken up by the cell

ix) The hybridomas are made by _____.

- a) Fusing T cells with myeloma cells
- b) Fusing T helper cells with myeloma cells
- c) Fusing B cells with myeloma cells
- d) Fusing B memory cells with myeloma cells

x) Cell-mediated immunity is carried out by _____. While humoral immunity is mainly carried out by _____.

- a) B cells/T cells
- b) Epitopes/Antigens
- c) T cells/B cells
- d) Antibodies/Antigens

xi) Type IV hypersensitivity is also called as _____.

- a) Immediate hypersensitivity
- b) Delayed hypersensitivity
- c) Cytotoxic hypersensitivity
- d) Immune complex hypersensitivity

xii) Which immunoglobulin is responsible for humoral sensitivity and histamine release?

- a) IgE
- b) IgM
- c) IgG
- d) IgA

xiii) _____ technique is used for amplification of RNA molecules

- a) Nested PCR
- b) Real time quantitative PCR
- c) Anchored PCR
- d) Reverse Transcription PCR (RT-PCR)

xiv) The transfer of genetic material from one bacterium to another via virus is called.....

- a) Transformation
- b) Conjugation
- c) Recombination
- d) Transduction

xv) Which of the following characteristics is not true of a plasmid?

- a) It is a circular piece of DNA.
- b) It is required for normal cell function.
- c) It is found in bacteria.
- d) It can be transferred from cell to cell.

xvi) The mutation which will not affect the length of a protein is _____.

- a) Nonsense mutation
- b) Missense mutation
- c) Frame shift mutation
- d) Reverse Mutation

xvii) Which of the following technique is suitable for identifying mRNA molecule in a sample

- a) Western blotting
- b) Southern blotting
- c) Eastern blotting
- d) Northern blotting

xviii) The chemical nature of humulin produced by rDNA technology is

- a) lipid
- b) protein
- c) polysaccharide
- d) nucleic acid

xix) A period during which the growth rate of cells gradually increases is known as _____.

- a) Lag phase
- b) Stationary phase
- c) Log phase
- d) Deceleration phase

xx) It is necessary to have _____ in the nutrient medium for the production of Vitamin B12.

- a) Manganese
- b) Phosphate
- c) Cobalt
- d) Calcium

Q. 2. Long Answers Questions (Answer 2 out of 3) (2 x 10) = 20

- i) Define enzyme immobilization. Elaborate different methods of enzyme immobilization and give their applications.
- ii) Describe the steps involved in r-DNA technology and its application in detail.
- iii) Illustrate immuno-blotting technique? Discuss in detail the different methods of immuno-blotting techniques.

Q. 3. Short Answers Questions (Answer 7 out of 9) (7 x 5) = 35

- i) Define the cloning vector. Explain the different types of cloning vectors.
- ii) Illustrate Hybridoma technology and give the role of HAT medium in monoclonal antibody production.
- iii) State the mutation and comment on different types of mutation.
- iv) Discuss the structure and functions of different types of MHC.
- v) Explain in detail the construction of a typical fermenter with a suitable diagram.
- vi) Define biosensor. Illustrate the components and types of biosensors.

vii) Write a note on PCR technique

viii) Elaborate in detail any four methods of gene transfer along with diagrams.

ix) Compare different types of hypersensitivity reactions.

-----END OF THE PAPER-----

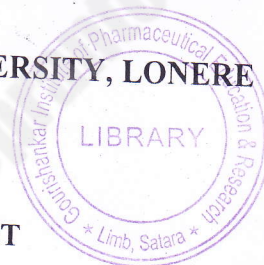
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 16/02/2023

Course: B. Pharmacy
Subject Name: Pharmaceutical Biotechnology
Max Marks: 75

Sem: VI
Subject Code: BP605T
Duration:



3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) =

- i) Write the applications of genetic engineering.
- ii) Enlist various enzymes used in rDNA technique with their functions.
- iii) Differentiate between Major Histocompatibility Complex(MHC) class I and class II molecules.
- iv) Define Immunity. Enlist types of immunity
- v) Give role of microbes in industry.
- vi) Define Antigen and Antibody.
- vii) Give the roles of Southern and Western blotting techniques.
- viii) Define: transduction and conjugation.
- ix) Comment on cloning vectors. Give their examples.
- x) Differentiate between prokaryotic and eukaryotic cells.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) =

- i) What is enzyme immobilization? Describe methods of enzyme immobilization along with their applications.
- ii) Explain in detail about the production and purification of monoclonal antibodies by hybridoma technology. Give their applications.
- iii) Explain the basic principle of rDNA technology. Write a detailed account on human insulin production by rDNA technology.

Q. 3. Short Answers (Answer 7 out of 9)

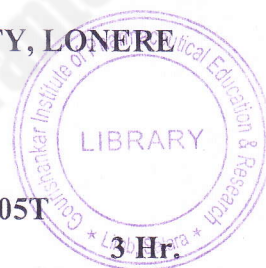
(7 x 5) =

- i) Write a short note on PCR.
- ii) Explain the types of mutation and mutants with suitable examples.
- iii) Elucidate the structure of immunoglobulins with neat labelled diagram.
- iv) Explain about the production of penicillin by fermentation technique.
- v) Write a note on dried human plasma and plasma substitutes.
- vi) Illustrate large scale production fermenter design with its various controls.
- vii) Describe type I hypersensitivity reaction with suitable diagram.
- viii) Illustrate the working principle and applications of biosensors.
- ix) Elaborate ELISA technique.

-----END OF THE PAPER-----

Course: B. Pharmacy
Subject Name: Pharmaceutical Biotechnology
Max Marks: 75

Sem: VI
Subject Code: BP605T
Duration: 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) What are plasmids? Give one example.
- ii) What is protein engineering?
- iii) Write the classification of hypersensitivity reactions.
- iv) Define terms: a) Transcription b) Translation
- v) Define antigen and antibody..
- vi) Differentiate the Vaccine and toxoid
- vii) Define vector. Write the features of a good vector.
- viii) What is gene therapy?
- ix) Enlist the enzymes used in DNA cloning
- x) Name four mutagenic agents.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Explain in details about production and purification of monoclonal antibodies by hybridoma technology
- ii) Describe the general method of recombinant DNA technology.
- iii) Explain different methods of enzyme immobilization with their advantages and disadvantages

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

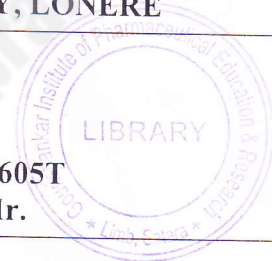
- i) What are biosensors? Write its principle and its functions.
- ii) Write a detailed note on cloning vectors.
- iii) What is PCR? Write the working principle of PCR.
- iv) Write in brief the Collection, Processing and Storage of whole human blood.
- v) Write in brief the southern blot technique and mention its applications.
- vi) Explain different types of mutations.
- vii) Write a note on microbial biotransformation in production of steroidal medicinal agents.
- viii) Describe the structure and functions of MHC.
- ix) Explain the production of Penicillin by fermentation technology with a neat labelled flow chart.

-----END OF THE PAPER-----

Date: 08/01/2024

Course : B. Pharmacy
Subject Name : Pharmaceutical Biotechnology
Max Marks : 75

Sem: VI
Subject Code : BP605T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Draw neat labelled diagram of structure of immunoglobulin.
- ii) What are mutants? Give their examples.
- iii) Differentiate between genetic organization of eukaryotes and prokaryotes.
- iv) Define biotechnology. Give its scope in pharmaceutical sciences.
- v) State the uses of microbes in industry.
- vi) What are the functions of DNA ligase and restriction endonucleases?
- vii) Write the importance of aeration and stirring in fermentation.
- viii) Define: Transformation and transduction.
- ix) Differentiate between cellular and humoral immunity.
- x) What is cold chain storage? State the storage conditions of vaccines.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What are biosensors? Explain the working and applications of biosensors in Pharmaceutical sciences.
- ii) Illustrate the principle of rDNA technology with neat labelled diagram. Give the detailed account of human insulin production by rDNA technology.
- iii) Explain the production and purification of monoclonal antibodies by hybridoma technology. Give their applications in pharmaceutical industry.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain in detail ELISA technique with neat labelled diagram.
- ii) Illustrate the design of large scale production fermenter and explain its various controls.
- iii) Explain the Collection and Storage of whole human blood. Extend the note on plasma substitutes.
- iv) Give comparative explanation of hypersensitivity reactions.
- v) Explain the structure and functions of Major Histocompatibility Complex (MHC).
- vi) Write a note on Polymerase Chain Reaction (PCR).
- vii) Explain the principle and methods of protein engineering.
- viii) Explain the methods of enzyme immobilization.
- ix) Write a note on cloning vectors in rDNA technology.

-----END OF THE PAPER-----

Branch: B. Pharm.

Subject: Pharmaceutical Biotechnology

Subject Code: BP605T

Date: 24/06/2024

Semester: VI

Marks: 75

Time: 3hrs

Instructions: i) All questions are compulsory.
ii) Figures to the right indicate full marks.
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Objective type questions. (Answer all the questions)

(2×10 =20 Marks)

- A] What are microbial transformation give two examples?
- B] Write the importance of transposomes.
- C] Describe the term protein engineering.
- D] What is hypersensitivity? Enlist their types.
- E] Distinguish between humoral mediated and cell mediated immunity.
- F] What are batch culture and continuous culture?
- G] What are plasma substitutes? Give its therapeutic uses.
- H] Enlist different types of vectors used in genetic engineering.
- I] Define the term biotechnology.
- J] Describe the structure of immunoglobulin with a neat labeled diagram.

Q. No.2 Long answers (Answer 2 out of 3)

(10×2 = 20 Marks)

- A] What is enzyme immobilization? Write in detail different methods of enzyme immobilization.
- B] Discuss in detail the different immuno-blotting techniques in detail (Northern, Western ELISA etc)
- C] Describe construction and working of a fermenter. Discuss briefly on the production of penicillin by fermentation technology.

Q. No.3 Short answers (Answer 7 out of 9)

(7× 5 = 35 Marks)

- A] Describe biosensors. Illustrate the components and types of biosensors.
- B] Explain briefly transformation, transduction and conjugation.
- C] Describe the PCR. Write its applications.
- D] Discuss the general method of preparation of live attenuated bacterial vaccines.
- E] Give a brief account of the vectors used in genetic engineering with a neat diagram.
- F] Write in brief the collection and storage of whole human blood.
- G] Explain about the monoclonal antibodies with a neat diagram. Write its applications.
- H] Discuss the production of Vitamin B12 by fermentation.
- I] Explain the different types of mutagenic agents.

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